STUDENT MANAGEMENT SYSTEM

CSE 221: Data Structure

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Abstract

This project is a console-based Student Management System developed using C++. It manages student records such as ID, name, age, and course. The system implements multiple data structures, including array, stack, and queue to perform operations efficiently. It allows users to add, delete, update, search, sort, and display student information along with undo functionality and basic statistics.

Introduction

Managing student data manually is inefficient and error-prone. This project demonstrates how data structures can be used to solve this problem efficiently. The system is menu-driven and performs all operations through a console interface.

The project mainly focuses on implementing real-world applications of:

- Arrays
- Stack
- Queue

Objectives

- To create a C++ system to store and manage flat information.
- To allow searching by location, rent range, or number of rooms.
- To add a Login & Registration system.
- To implement add/update/delete flat details.
- To store all data using file handling.
- To provide a simple console-based interface.

Data Structures Used

5.1 Array (Main Storage)

An array of structures is used to store student records.

- Stores all student data
- Provides direct access using the index
- Simple and efficient for fixed-size data

5.2 Stack (Undo Functionality)

A stack is implemented using a linked list.

Used for:

- Insert undo
- Delete undo
- Update tracking

5.3 Queue (Report System)

Queue is implemented using linked list.

Used for:

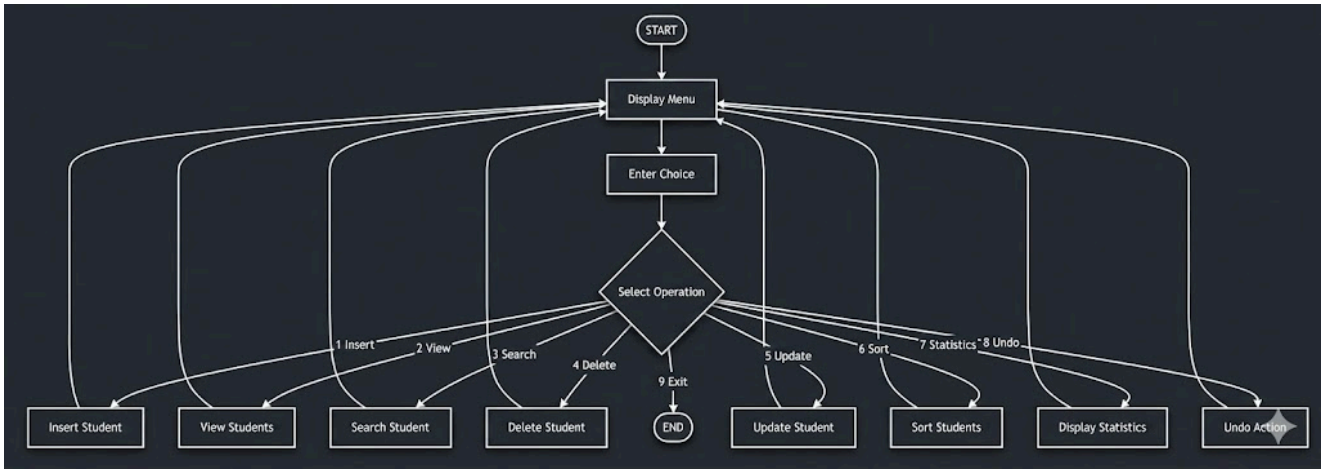
- Statistics report handling

System Features

1. Add Student Record
2. View All Students
3. Search by ID
4. Search by Name
5. Delete Student Record
6. Update Student Record
7. Sort Students by Name (Bubble Sort)
8. Display Statistics (Average Age)
9. Undo Last Action (Stack-based)

System Design

Flow of the program



Algorithms Used

8.1 Search Algorithm

Linear search is used for searching student records.

8.2 Sorting Algorithm

Bubble Sort is used to sort students by name.

8.3 Undo Mechanism

Stack stores previous operations and restores the last action.

Functional Description

Add Student

- Input ID, Name, Age, Course
- Check duplicate ID
- Store in array
- Save action in stack

Delete Student

- Find by ID
- Remove from array
- Save in stack

Update Student

- Find student
- Save old data
- Update new values

Statistics

- Calculate total students
- Compute the average age
- Use a queue for report handling

Advantages

1. Simple and efficient system
2. Demonstrates real-life data structure usage
3. Supports the undo feature
4. Fast operations using arrays

Limitations

1. Fixed-size array (100 students)
2. No database support
3. Console-based interface only
4. No graphical UI

Future Improvements

- Add database (MySQL)
- Develop a GUI version
- Increase scalability
- Add a login system
- Export data to files

Conclusion

This project successfully demonstrates the implementation of a Student Management System using multiple data structures in C++. It improves understanding of arrays, stacks, and queues through practical implementation and real-world problem solving.